

Modul Praktikum Manajemen Basis Data

Konsep DBMS dan MySQL Dasar

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Teknik Informatika
Universitas Muhammadiyah Bandung
2024

Mata Kuliah: Manajemen Basis Data

Durasi: 170 menit

Tujuan:

- Mahasiswa memahami dan dapat menggunakan perintah dasar MySQL.
- Mahasiswa mampu menerapkan konsep database sederhana untuk memanipulasi data.

Materi dan Latihan

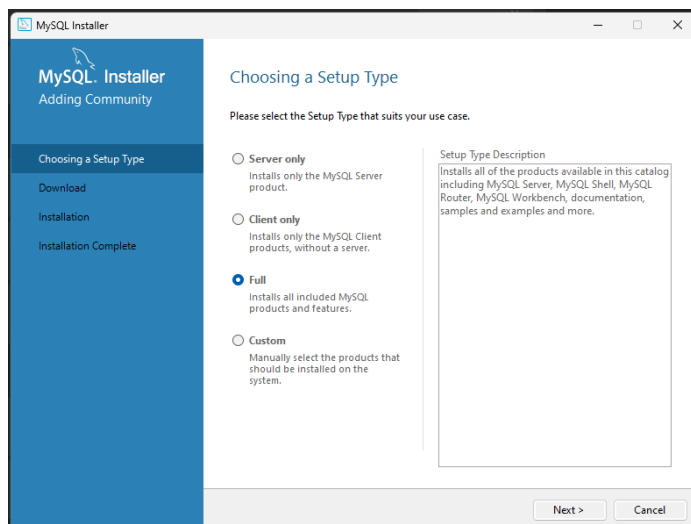
A. Pendahuluan

Pada praktikum ini, Anda akan membuat sebuah basis data sederhana untuk mengelola informasi mahasiswa. Praktikum mencakup:

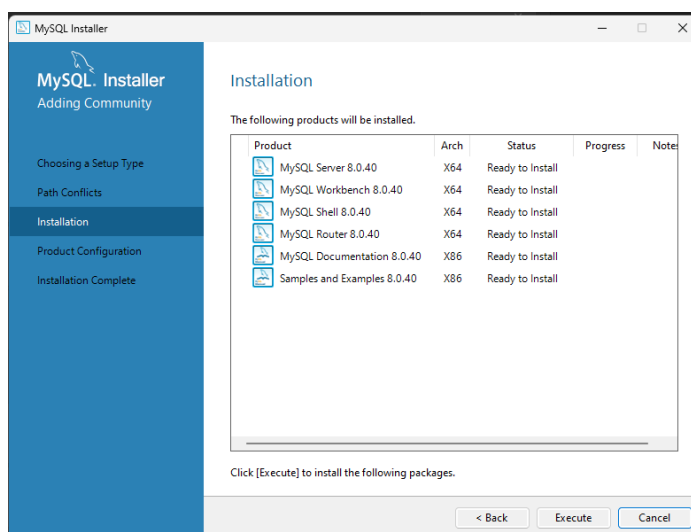
1. Membuat database dan tabel.
2. Mengelola data menggunakan perintah dasar MySQL (CRUD).
3. Studi kasus sederhana.

B.Instalasi MySQL

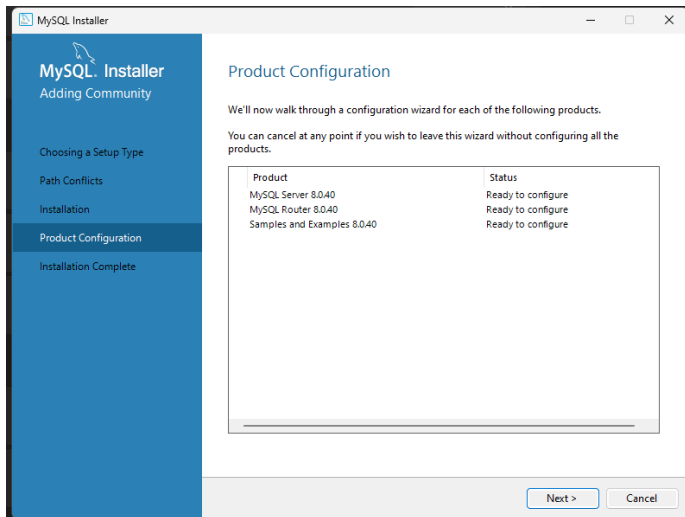
1.



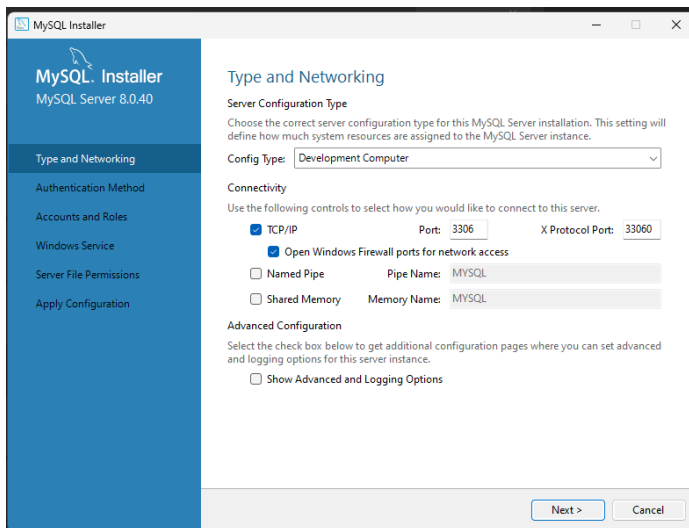
2.



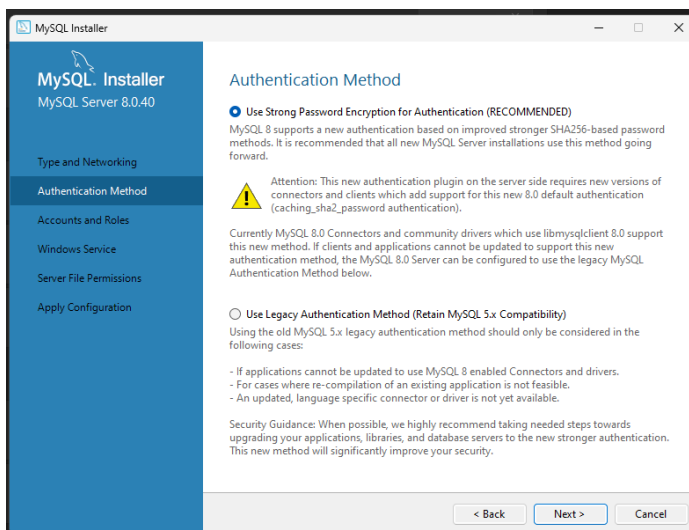
3.



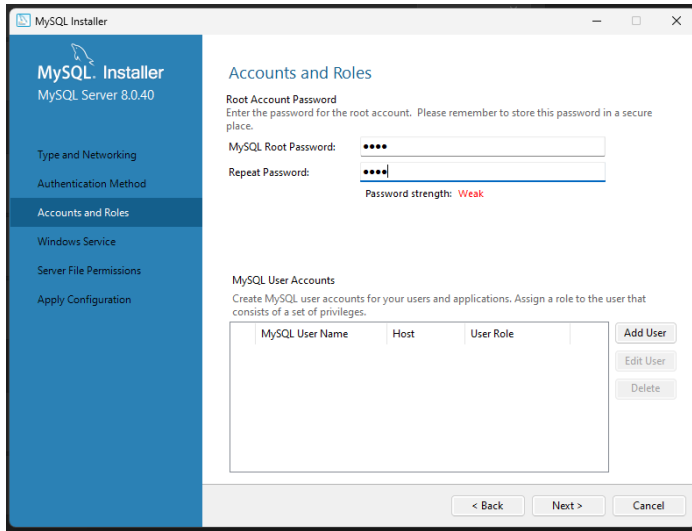
4.



5.



6.



The screenshot shows the 'Accounts and Roles' step of the MySQL 8.0.40 installer. The left sidebar lists the installation steps: Type and Networking, Authentication Method, Accounts and Roles (selected), Windows Service, Server File Permissions, and Apply Configuration. The main area is titled 'Accounts and Roles' and contains two sections. The first section, 'Root Account Password', prompts the user to enter a password for the root account. It includes fields for 'MySQL Root Password' and 'Repeat Password', both masked with dots. Below these fields, the password strength is indicated as 'Weak'. The second section, 'MySQL User Accounts', instructs the user to create MySQL user accounts. It features a table with columns for 'MySQL User Name', 'Host', and 'User Role'. To the right of the table are buttons for 'Add User', 'Edit User', and 'Delete'. At the bottom of the window are navigation buttons: '< Back', 'Next >', and 'Cancel'.

MySQL Installer
MySQL Server 8.0.40

Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration

Accounts and Roles

Root Account Password
Enter the password for the root account. Please remember to store this password in a secure place.

MySQL Root Password:

Repeat Password:

Password strength: **Weak**

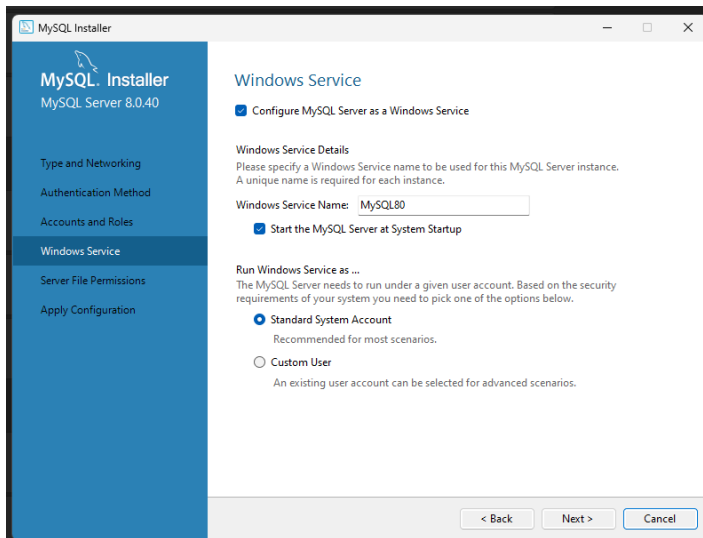
MySQL User Accounts
Create MySQL user accounts for your users and applications. Assign a role to the user that consists of a set of privileges.

MySQL User Name	Host	User Role
-----------------	------	-----------

Add User
Edit User
Delete

< Back Next > Cancel

7.



The screenshot shows the 'Windows Service' step of the MySQL 8.0.40 installer. The left sidebar is the same as in the previous step, with 'Windows Service' now selected. The main area is titled 'Windows Service' and contains several options. At the top, there is a checkbox labeled 'Configure MySQL Server as a Windows Service', which is checked. Below this, the 'Windows Service Details' section asks the user to specify a Windows Service name. A text box contains 'MySQL80'. There is also a checkbox labeled 'Start the MySQL Server at System Startup', which is checked. The 'Run Windows Service as ...' section explains that the MySQL Server needs to run under a given user account. It offers two options: 'Standard System Account' (selected) and 'Custom User'. The 'Standard System Account' option is noted as 'Recommended for most scenarios.' The 'Custom User' option is noted as 'An existing user account can be selected for advanced scenarios.' At the bottom of the window are navigation buttons: '< Back', 'Next >', and 'Cancel'.

MySQL Installer
MySQL Server 8.0.40

Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration

Windows Service

☒ Configure MySQL Server as a Windows Service

Windows Service Details
Please specify a Windows Service name to be used for this MySQL Server instance. A unique name is required for each instance.

Windows Service Name:

☒ Start the MySQL Server at System Startup

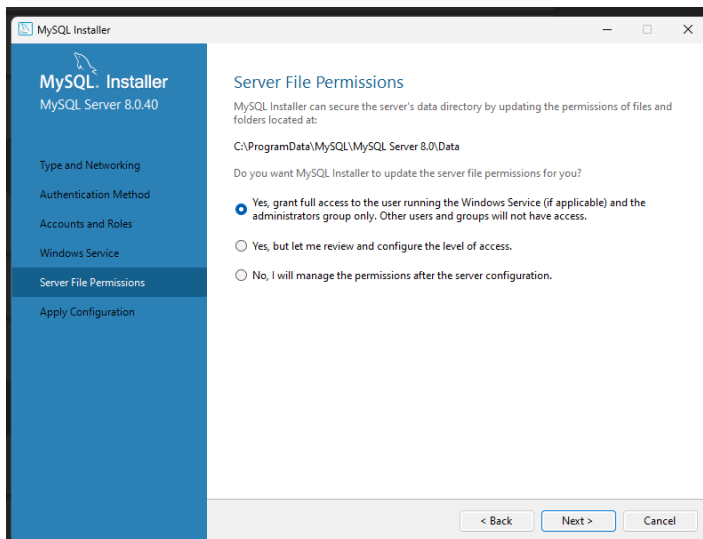
Run Windows Service as ...
The MySQL Server needs to run under a given user account. Based on the security requirements of your system you need to pick one of the options below.

☒ Standard System Account
Recommended for most scenarios.

☐ Custom User
An existing user account can be selected for advanced scenarios.

< Back Next > Cancel

8.



The screenshot shows the 'Server File Permissions' step of the MySQL 8.0.40 installer. The left sidebar is the same as in the previous steps, with 'Server File Permissions' now selected. The main area is titled 'Server File Permissions' and explains that the installer can secure the server's data directory by updating the permissions of files and folders located at: 'C:\ProgramData\MySQL\MySQL Server 8.0\Data'. It then asks the user: 'Do you want MySQL Installer to update the server file permissions for you?'. There are three radio button options: 'Yes, grant full access to the user running the Windows Service (if applicable) and the administrators group only. Other users and groups will not have access.' (selected), 'Yes, but let me review and configure the level of access.', and 'No, I will manage the permissions after the server configuration.' At the bottom of the window are navigation buttons: '< Back', 'Next >', and 'Cancel'.

MySQL Installer
MySQL Server 8.0.40

Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration

Server File Permissions

MySQL Installer can secure the server's data directory by updating the permissions of files and folders located at:

C:\ProgramData\MySQL\MySQL Server 8.0\Data

Do you want MySQL Installer to update the server file permissions for you?

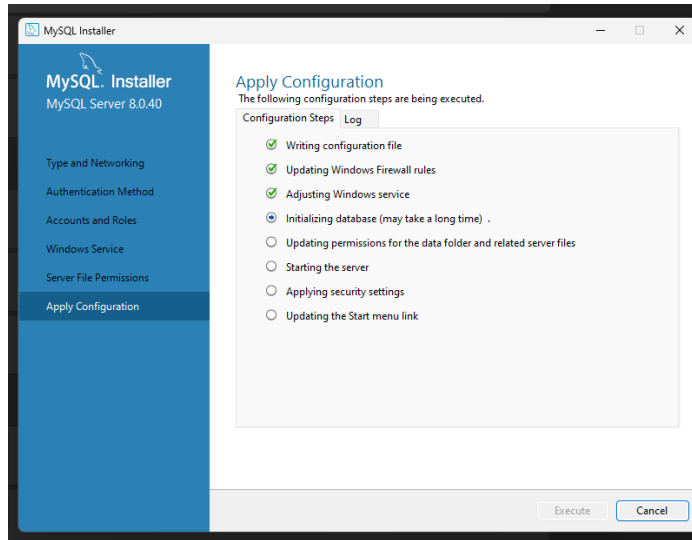
☒ Yes, grant full access to the user running the Windows Service (if applicable) and the administrators group only. Other users and groups will not have access.

☐ Yes, but let me review and configure the level of access.

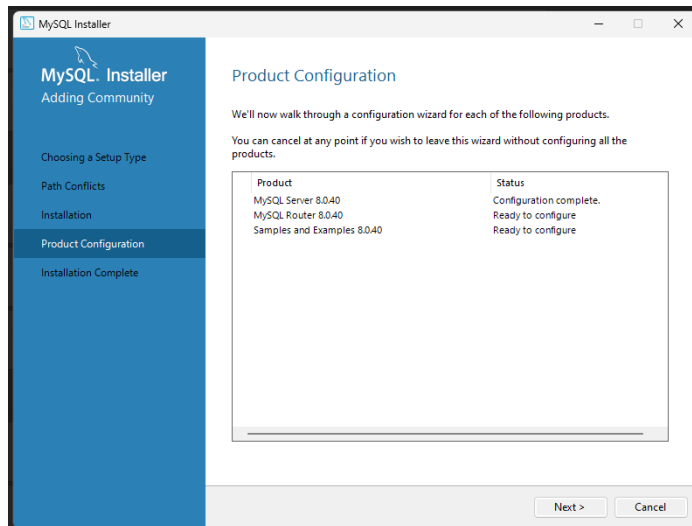
☐ No, I will manage the permissions after the server configuration.

< Back Next > Cancel

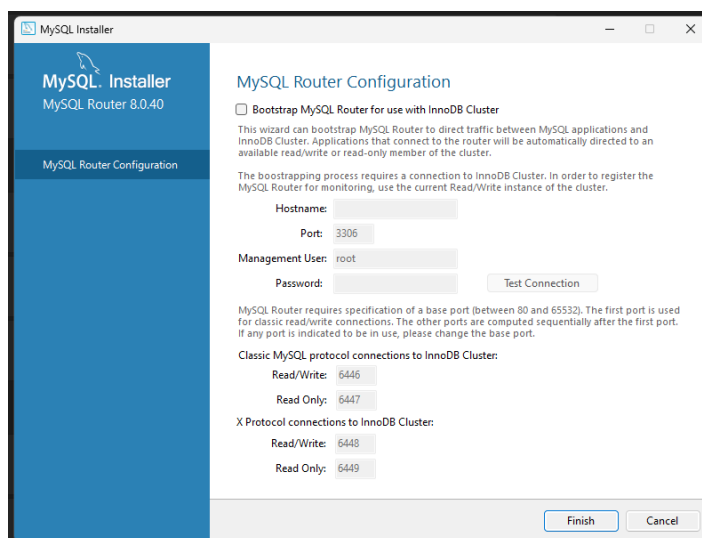
9.



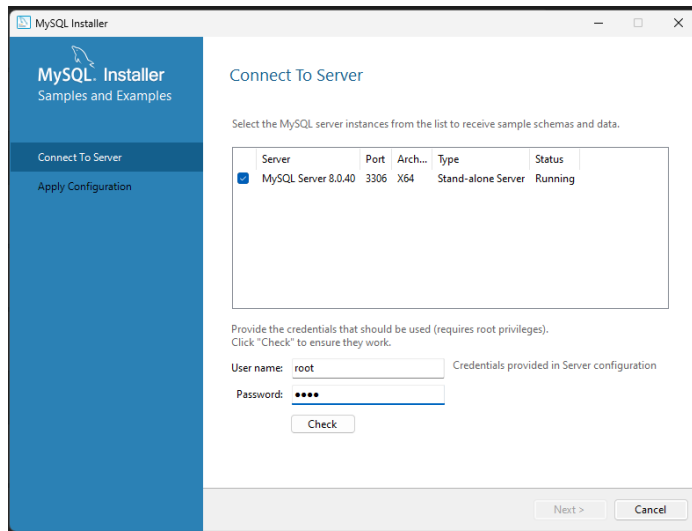
10.



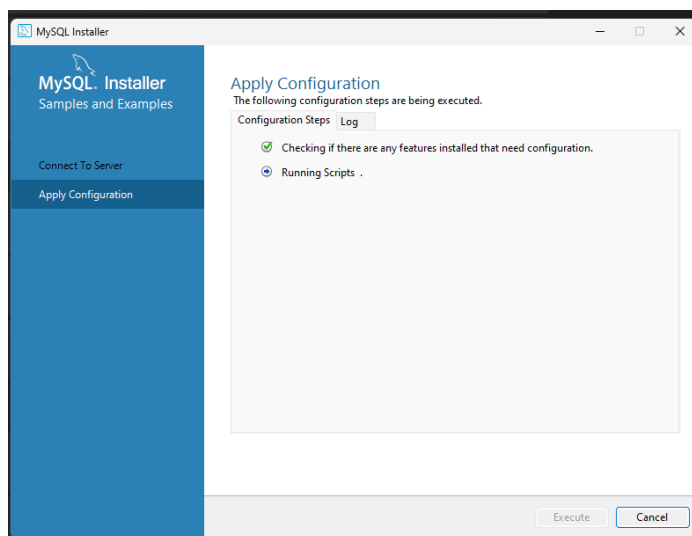
11.



12.



13.



C. Perintah Dasar MySQL

1. Membuat Database

Latihan :

- Buat database baru bernama universitas.

Perintah SQL:

```
create database Universitas;

use Universitas;
```

Screenshot hasil running query

2. Membuat Tabel

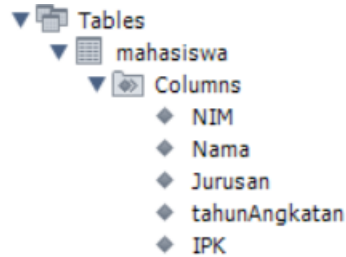
Latihan:

- Buat tabel mahasiswa dengan kolom berikut:
 - nim (Nomor Induk Mahasiswa) sebagai Primary Key tipe varchar.
 - nama (Nama Mahasiswa) tipe varchar.
 - jurusan (Jurusan Mahasiswa) tipe varchar.
 - angkatan (Tahun Angkatan) tipe Year.
 - ipk (Indeks Prestasi Kumulatif) tipe decimal.

Perintah SQL:

```
create table Mahasiswa(  
  NIM varchar (50) primary key,  
  Nama varchar (50),  
  Jurusan varchar (50),  
  tahunAngkatan year,  
  IPK float  
);
```

Screenshot hasil running query



3. Menambahkan Data ke Tabel

Latihan:

- Tambahkan 3 data mahasiswa berikut ke tabel mahasiswa.

Nim	Nama	jurusan	Angkatan	ipk
12345	Ali Hasan	Informatika	2021	3.75
12346	Budi Santoso	Sistem Informasi	2022	3.60
12347	Citra Dewi	Teknik Komputer	2020	3.85

Perintah SQL:

```

insert into Mahasiswa(NIM,Nama,Jurusan,tahunAngkatan,IPK)
values ('12345','Ali Hasan','Informatika',2021,3.75),
('12346','Budi Santoso','Sistem Informasi',2022,3.60),
('12347','Citra Dewi','Teknik Komputer',2022,3.85);
  
```

Screenshot hasil running query

23	14.43.15	create table Mahasiswa(NIM varchar (50) primary key, Nama varchar (50), Jurusan varchar (50), tahunAngkatan year, IPK float)	0 row(s) affected	0.094 sec
24	14.43.28	insert into Mahasiswa(NIM,Nama,Jurusan,tahunAngkatan,IPK) values ('12345','Ali Hasan','Informatika',2021,3.75), ('12346','Budi Santoso','Sistem Informasi',2022,3.60), ('12347','Citra Dewi','Teknik Komputer',2022,3.85)	3 row(s) affected	0.032 sec

4. Membaca Data dari Tabel

Latihan:

- Tampilkan semua data mahasiswa.
- Tampilkan mahasiswa dari jurusan Informatika
- Tampilkan mahasiswa dengan IPK lebih dari 3.7

Perintah SQL:

- Menampilkan semua data

```
select * from Mahasiswa;
```

Screenshot hasil running query

NIM	Nama	Jurusan	tahunAngkatan	IPK
12345	Ali Hasan	Informatika	2021	3.75
12346	Budi Santoso	Sistem Informasi	2022	3.6
12347	Citra Dewi	Teknik Komputer	2022	3.85

- Menampilkan mahasiswa dari jurusan "Informatika"

```
select * from Mahasiswa where Jurusan = 'Informatika';
```

Screenshot hasil running query

	NIM	Nama	Jurusan	tahunAngkatan	IPK
▶	12345	Ali Hasan	Informatika	2021	3.75

- Menampilkan mahasiswa dengan IPK lebih dari 3.7

```
select * from Mahasiswa where IPK > 3.7;
```

Screenshot hasil running query

	NIM	Nama	Jurusan	tahunAngkatan	IPK
▶	12345	Ali Hasan	Informatika	2021	3.75
	12347	Citra Dewi	Teknik Komputer	2022	3.85

5. Memperbarui Data

Latihan:

- Perbarui IPK mahasiswa dengan NIM '12346' menjadi 3.8.

Perintah SQL:

```
update Mahasiswa set IPK = 3.8 where NIM = "12346";
select * from Mahasiswa where NIM = "12346";
```

Screenshot hasil running query

	NIM	Nama	Jurusan	tahunAngkatan	IPK
▶	12346	Budi Santoso	Sistem Informasi	2022	3.8
*	NULL	NULL	NULL	NULL	NULL

6. Menghapus Data

Latihan:

- Hapus data mahasiswa dengan Nama 'Citra Dewi'.

Perintah SQL

```
delete from Mahasiswa where NIM = "12347";
select * from Mahasiswa;
```

Screenshot hasil running query

	NIM	Nama	Jurusan	tahunAngkatan	IPK
▶	12345	Ali Hasan	Informatika	2021	3.75
	12346	Budi Santoso	Sistem Informasi	2022	3.8
*	NULL	NULL	NULL	NULL	NULL

7. Mengubah struktur tabel

Latihan :

- Tambahkan kolom tanggal lahir pada tabel mahasiswa.
- Masukkan nilai tanggal lahir pada setiap data di tabel mahasiswa

Perintah SQL :

```
alter table Mahasiswa add tanggal_lahir date ;  
update Mahasiswa set tanggal_lahir = "1994-08-12" where NIM = "12345";  
update Mahasiswa set tanggal_lahir = "1994-08-10" where NIM = "12346";  
select * from Mahasiswa;
```

Screenshot hasil running query

	NIM	Nama	Jurusan	tahunAngkatan	IPK	tanggal_lahir
▶	12345	Ali Hasan	Informatika	2021	3.75	1994-08-12
	12346	Budi Santoso	Sistem Informasi	2022	3.8	1994-08-10
*	NULL	NULL	NULL	NULL	NULL	NULL